Assignment 3 Hint

Clamp to [0, 255]

```
output = (uint8_t)min(max((int)roundf(sum), 0), 255);
```

Note: `roundf()` is required to match the reference image.

Sobel Kernel (global memory version)
Clamp-to-edge helper

```
__global__ void sobelKernel(
    const uint8_t* __restrict__ in,
    uint8_t*       __restrict__ out,
    int width, int height)
{
    int x = blockIdx.x * blockDim.x + threadIdx.x;
    int y = blockIdx.y * blockDim.y + threadIdx.y;

    if (x ≥ width || y ≥ height) return;

    // Clamp helper — returns pixel value at (px, py) with edge clamping.
    auto px = [&](int px, int py) → float {
        px = min(max(px, 0), width  - 1);
        py = min(max(py, 0), height - 1);
        return (float)in[py * width + px];
    };

    // 3x3 neighbourhood
    float p00 = px(x-1,y-1), p10 = px(x,y-1), p20 = px(x+1,y-1);
    float p01 = px(x-1,y  ), p11 = px(x,y),   p21 = px(x+1,y  );
    float p02 = px(x-1,y+1), p12 = px(x,y+1), p22 = px(x+1,y+1);
```

Assignment 3 Hint

Implement one kernel at a time.
A correct implementation should pass with MaxDiff = 0.
For example, after completing stage 2:

```
python3 scripts/check_outputs.py --output output \
--reference data/expected_output/stage-2
```

```
Image                                          Stage         MaxDiff   MeanDiff   Result
------------------------------------------------------------------------------------------
  img_01_gradient_h_out.pgm                      2                 0       0.00     PASS
  img_02_gradient_v_out.pgm                      2                 0       0.00     PASS
  img_03_gradient_diag_out.pgm                   2                 0       0.00     PASS
  img_04_checkerboard_out.pgm                    2                 0       0.00     PASS
  img_05_circles_out.pgm                         2                 0       0.00     PASS
  img_06_gaussian_blob_out.pgm                   2                 0       0.00     PASS
  img_07_blobs_out.pgm                           2                 0       0.00     PASS
  img_08_sine_grid_out.pgm                       2                 0       0.00     PASS
  img_09_noise_uniform_out.pgm                   2                 0       0.00     PASS
  img_10_noise_gaussian_out.pgm                  2                 0       0.00     PASS
  img_11_stripes_v_out.pgm                       2                 0       0.00     PASS
  img_12_stripes_h_out.pgm                       2                 0       0.00     PASS
  img_13_gradient_noisy_out.pgm                  2                 0       0.00     PASS
  img_14_dark_lowcontrast_out.pgm                2                 0       0.00     PASS
  img_15_bright_highkey_out.pgm                  2                 0       0.00     PASS
  img_16_radial_out.pgm                          2                 0       0.00     PASS
  img_17_angular_out.pgm                         2                 0       0.00     PASS
  img_18_salt_pepper_out.pgm                     2                 0       0.00     PASS
  img_19_concentric_squares_out.pgm              2                 0       0.00     PASS
  img_20_mixed_out.pgm                           2                 0       0.00     PASS
------------------------------------------------------------------------------------------

Results: 20 passed, 0 failed, 0 missing reference
Tolerance: ±2 intensity levels
```